SIM2001 Advanced Calculus : Tutorial 2

1. (13.4.41)

   Assume that $w = f(s^2 + t^2)$ and $f'(x) = e^x$. Find $\dfrac{\partial w}{\partial s}$ and $\dfrac{\partial w}{\partial t}$.

2. (13.4.51)

   Show that if $w = f(u, v)$ satisfies the **Laplace equation** $f_{uu} + f_{vv} = 0$ and if $u = \dfrac{x^2 - y^2}{2}$ and $v = xy$, then w satisfies the Laplace equation $w_{xx} + w_{yy} = 0$.

3. Find the gradient of the function at the given point. Then sketch the gradient together with the level curve that passes through the point.

   (13.5.3) $g(x, y) = xy^2, \quad (2, -1)$
   (13.5.5) $f(x, y) = \sqrt{2x + 3y}, \quad (-1, 2)$

4. Find ∇f at the given point.

   (13.5.7) $f(x, y, z) = x^2 + y^2 - 2z^2 + z \ln x, \quad (1, 1, 1)$

5. Find the derivative of the function at P_0 in the direction of $\underset{\sim}{u}$.

   (13.5.13) $g(x, y) = \dfrac{x - y}{xy + 2}, \quad P_0(1, -1), \quad \underset{\sim}{u} = 12\underset{\sim}{i} + 5\underset{\sim}{j}$
   (13.5.17) $g(x, y, z) = 3e^x \cos yz, \quad P_0(0, 0, 0), \quad \underset{\sim}{u} = 2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k}$

6. Find the directions in which the function increases and decreases most rapidly at P_0. Then find the derivative of the function in this direction.

   (13.5.21) $f(x, y, z) = \dfrac{x}{y} - yz, \quad P_0(4, 1, 1)$

7. (13.5.29)
   Let $f(x, y) = x^2 - xy + y^2 - y$. Find the directions $\underset{\sim}{u}$ and the values of $D_{\underset{\sim}{u}} f(1, -1)$ for which

   (a) $D_{\underset{\sim}{u}} f(1, -1)$ is largest
   (b) $D_{\underset{\sim}{u}} f(1, -1)$ is smallest
   (c) $D_{\underset{\sim}{u}} f(1, -1) = 0$
   (d) $D_{\underset{\sim}{u}} f(1, -1) = 4$
   (e) $D_{\underset{\sim}{u}} f(1, -1) = -3$

8. Find equations for the

   (a) tangent plane and
   (b) normal line at the point P_0 on the given surface.

   (13.6.3) $2z - x^2 = 0, \quad P_0(2, 0, 2)$
   (13.6.5) $\cos \pi x - x^2 y + e^{xz} + yz = 4, \quad P_0(0, 1, 2)$

9. Find parametric equations for the line tangent to the curve of intersection of the surfaces at the given point.

   (13.6.15) Surface : $x^2 + 2y + 2z = 4, \quad y = 1;$ Point $(1, 1, 1/2)$
   (13.6.19) Surface : $x^3 + 3x^2y^2 + y^3 + 4xy - z^2 = 0, \quad x^2 + y^2 + z^2 = 11;$ Point $(1, 1, 3)$

10. (13.6.23)
    By about how much will $g(x, y, z) = x + x\cos z - y\sin z + y$ changes if the point $P(x, y, z)$ moves from $P_0(2, -1, 0)$ a distance of $ds = 0.2$ unit toward the point $P_1(0, 1, 2)$?

11. (13.6.25)
    Suppose that the Celsius temperature at the point (x, y) in the xy-plane is $T(x, y) = x\sin 2y$ and that distance in the xy-plane is measured in meters. A particle is moving **clockwise** around the circle of radius 1 m centered at the origin at the constant rate of 2 m/sec.

    (a) How fast is the temperature experienced by the particle changing in degrees Celsius per meter at the point $P(1/2, \sqrt{3}/2)$?

    (b) How fast is the temperature experienced by the particle changing in degrees Celsius per second at P?

12. Find the linearization $L(x, y)$ of the function $f(x, y)$ at P_0. Then find an upper bound for the magnitude $|E|$ of the error in the approximation $f(x, y) \approx L(x, y)$ over the rectangle R.

    (13.6.35) $f(x, y) = x^2 - 3xy + 5$ at $P_0(2, 1)$; $\quad R : |x - 2| \le 0.1,\ |y - 1| \le 0.1$

    (13.6.39) $f(x, y) = e^x \cos y$ at $P_0(0, 0)$; $\quad R : |x| \le 0.1,\ |y| \le 0.1$
    (Use $|e^x| \le 1.11$, $|\cos y| \le 1$ in estimating E.)